GIANLUIGI VITALE

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RESEARCH INTERESTS

Machine learning systems, LLM serving infrastructure, accelerator portability, TPU kernels and numerical behavior, ML benchmarking, inference reliability, and LLM safety evaluation.

EDUCATION

B.Sc. in Computer Engineering — Universitas Mercatorum, Rome, Italy **Expected March 2027**

MUR-accredited university · GPA: 29.35/30 (Italian scale) · 3.87/4.0 (WES-converted) · expected final grade 110/110 cum laude — all coursework completed; thesis in progress

Thesis: DSProfiler — Static Analysis and Machine Learning for Automated Detection of Data Structure Performance Anti-patterns in Python

Supervisor: Prof. Rocco Pietrini

Full Stack Web Development Bootcamp — *Boolean SRL (700 hours)* *Jan.–Jun. 2020*

PUBLICATIONS

G. Vitale, "Decomposing One Professional-Framing Pipeline: Which Components Shift LLM Safety Boundaries?," **NeurIPS 2026**, main track. Accepted; to appear. Paper: *https://openreview.net/forum?id=1UvkQdlJOO*

G. Vitale, "DriftBench: Measuring and Predicting Infrastructure Drift in LLM Serving Systems," **MLSys 2026**, Bellevue, WA, May 2026. Oral presentation.
Paper: *https://proceedings.mlsys.org/paper_files/paper/2026/hash/ea0b5818ae9255ee1fb1e3b4442d2ffe-Abstract-Conference.html*
Artifact (ACM badges: Available, Functional, Reproduced): DOI *10.5281/zenodo.19361066*
Talk: *https://youtu.be/KnsVj5DdIEo*

HONORS & AWARDS

Distinguished Artifact Evaluator Award — *SOSP 2026 Artifact Evaluation Committee* *2026*

Oral Presentation — *MLSys 2026 (DriftBench)* *2026*

ACM Artifact Badges: Available, Functional, Reproduced — *MLSys 2026 (DriftBench)* *2026*

RESEARCH EXPERIENCE

- **DriftBench and TPU validation:** Built a measurement and prediction framework spanning 236,985 prompt-response pairs across 105 configurations. The Portability Risk Index generalizes to unseen hardware ($R^2 = 0.909$). Extended the result to TPU v5e/v6e, where predictability held across a new numerical path ($R^2 = 0.72–0.96$).
- **Factorial ablation of jailbreak framing (NeurIPS 2026):** Applied a pre-specified 2×2×2 factorial design to the STF professional-framing pipeline, within an evaluation of approximately 31,900 trials across nine LLMs. In the primary study, persona adoption was the main driver of acceptance (OR = 6.5), terminology substitution the main driver of response depth, and moral justification showed no detectable positive effect.
- **DeepSeek V4-Flash on TPU:** Completed a 32-chip v4-64 bring-up, validating every attention mode with both prefill and decode operational. Equivalent support has since landed in vLLM TPU inference through parallel development in the wider ecosystem.
- **DSA kernel for TPU:** Developed a working implementation that is bit-exact against the reference through 8K tokens and runs through 256K. Work in progress; it underpins the GLM-5.3 port, now running on TPU v4.

- **Bit-level TPU MXU characterization:** Characterized TPU v5e/v6e numerical behavior and built a CPU simulator that reproduces real Llama-3.1-8B matrix multiplications bit-for-bit on a native pass, enabling deterministic study of TPU arithmetic.
- **Long-context genomics with ICAR-CNR Italy:** Collaborating on whether long-context DNA models encode linkage disequilibrium, combining ML systems work with genomics expertise at Italy's National Research Council.

ACADEMIC SERVICE

Program Committee Member — *MLSys 2027* *2026–2027*
Reviewing submissions for the MLSys 2027 technical program.

Ethics Reviewer — *NeurIPS 2026* 2026
Reviewed submissions for ethical concerns, potential societal harms, research-integrity risks, and compliance with the conference's ethics guidelines.

Artifact Evaluation Committee Member — *SOSP 2026* *2026*
Evaluated research artifacts in memory-efficient model serving and memory-safety analysis for GPU kernels in LLM inference systems. Assessed availability, functionality, reproducibility, and consistency with the associated papers' experimental claims. Received the Distinguished Artifact Evaluator Award.

Artifact Evaluation Committee Member — *MLSys 2026* *2026*
Evaluated four accepted submissions spanning attention-kernel optimization, ML profiling infrastructure, GPU memory disaggregation, and distributed inference. Reproduced experimental results on rented 8×H100 clusters (RunPod) and AMD MI350X infrastructure (AMD University Program), assessing availability, functionality, and consistency with each paper's claims.

PROFESSIONAL EXPERIENCE

Systems Engineer — Archivio di Stato di Pistoia (Italian State Archive), Italian Ministry of Culture
Apr. 2024–Present

Designed and deployed a retrieval-augmented generation system over 10,000 internal administrative documents, delivering source-cited answers to staff queries. Built an AI-assisted public consultation interface giving staff and citizens natural-language access to the full inventory database.

Web Developer — *Informamuse S.R.L.* *Oct. 2020–Apr. 2021*
Developed websites and web applications using HTML, CSS, JavaScript, PHP, Laravel, Vue.js, and MySQL. Managed projects end-to-end from design through deployment.

TECHNICAL SKILLS

Languages: Python, JavaScript, PHP, SQL, HTML/CSS, Bash

Frameworks: PyTorch, JAX, Pallas, Hugging Face Transformers

ML/Systems: vLLM, tpu-inference, SGLang, TensorRT-LLM, CUDA, Docker

Infrastructure: Google Cloud TPU (v4, v5e, v6e), Linux administration, networking, cloud GPU provisioning (RunPod, AMD HPC/Chameleon Cloud), CI/CD

Research Tools: LaTeX, Git, Jupyter, scipy, statsmodels

COMPUTATIONAL RESOURCES

- Google Cloud TPU Research Cloud (TRC) - active since May 2026. Current 320-chip allocation: 32 on-demand v4 plus 288 spot chips across v4, v5e, and v6e in US and EU zones.
- AMD University Program account (active since March 2026)
- Chameleon Cloud allocation (2,000 SUs)
- RunPod GPU cluster experience (H100, H200, B200, MI300X configurations)

LANGUAGES

Italian (native), **English** (C1 — academic proficiency, scientific publications)

CONTACTS & PROFILES

Google Scholar: https://scholar.google.com/citations?hl=en&user=x4K6NWUAAAAJ | ORCID: https://orcid.org/0009-0007-6740-2695

OpenReview: https://openreview.net/profile?id=~Gianluigi_Vitale1 | Portfolio: https://gianluigivitale.github.io/portfolio/

GitHub: https://github.com/GianluigiVitale | LinkedIn: https://www.linkedin.com/in/gianluigi-vitale615